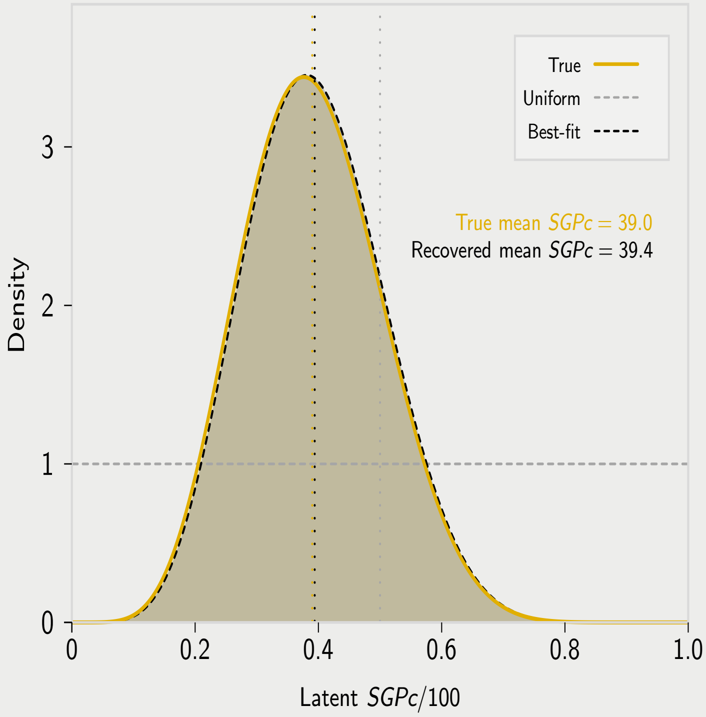
C. Recover inferred latent SGPc density $P_S \sim \hat{H}_S$

Report density/summary: mean $SGPc = 100\hat{m}$



The output of Panels B1–B2 is a subgroup-specific *estimated growth regime* $\hat{H}_S$, where $\hat{H}_S$ is a cumulative distribution function (CDF) on $[0, 1]$ for a latent conditional-percentile random variable P_S (SGPc on the 0–1 scale):

$$\hat{H}_S(p) \;=\; \Pr(P_S \le p), \qquad 0 \le p \le 1.$$

In words, $\hat{H}_S$ describes how subgroup S allocates conditional ranks *within* the baseline dependence template $F_0(\cdot \mid u)$.

Panel C displays the *distribution of* P_S (typically via its density $f_{\hat{H}_S}(p) = \hat{H}_S'(p)$ when $\hat{H}_S$ is smooth), not the CDF itself. Thus the "Best-fit" curve is the density corresponding to the fitted CDF (i.e., $P_S \sim \hat{H}_S$). The dashed grey "Uniform" reference corresponds to the uniform regime $H(p) = p$ (equivalently $P \sim \mathrm{Unif}(0, 1)$), which places no tilt on conditional percentiles relative to the baseline kernel.

Because we restrict the candidate regime class $\mathscr{H}$ to the Beta family in Panel B1, the fitted regime implies $P_S \sim \mathrm{Beta}(\kappa m, \kappa(1-m))$: the mean $m = \mathbb{E}[P_S]$ sets the regime center (reported as mean SGPc = $100m$), and κ controls concentration of latent SGPc.

The vertical reference lines mark mean SGPc under the uniform regime (50), the generating "True" regime (39.0), and the recovered best-fit regime (39.4). In applications, the primary outputs are $\hat{H}_S$ (with uncertainty bands) and summary functionals such as mean/median SGPc and tail/bucket probabilities.